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Enhanced reliability of aluminum–sulfur batteries with cost-effective ionic liquid electrolyte and sulfur/graphite cathode†

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The low coulombic efficiency and mild conductivity have impeded the commercialization of sulfur-based batteries despite pairing with high energy density and low-cost aluminum anodes. In this study, we investigated cost-effective electrolyte confinement of the sulfur cathode to an Al/S@Gf battery, which employed a sulfur-graphite (S@Gf) cathode, an aluminum metal anode, and 1-ethyl-3-methyl imidazolium chloride (EMImCl):AlCl₃ as an electrolyte. The Al/EMImCl:AlCl₃/S@Gf battery exhibited remarkable performance in terms of rate capability and cycle life, delivering a high capacity of 1281 mA h g^{−1} at a current density of 300 mA g^{−1} with 90% coulombic efficiency (CE); additionally, it was stable at 800 mA g^{−1} withdrawing current. By employing an embedding technique to incorporate sulfur particles onto the graphite, we emulated previously observed glitches, such as capacity degradation and poor conductivity resulting in the accumulation of insoluble discharge products hindering ion diffusion. We demonstrated the potential of graphite to enhance the capability of holding sulfur for creating cells with repeatable, stable, and long-lasting charge storage features. This paper elucidates the interactions between carbon, sulfur, and aluminum electrolytes, and sheds light on their respective roles in the fundamental charge-storage process of aluminum–sulfur batteries.

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1. Introduction

The expansion of cost-effective electrochemical energy storage solutions beyond the current state-of-the-art Li-ion batteries is of utmost importance for the battery research community, particularly in the context of improving grid-scale energy storage capabilities.^{1–3} Recent scientific investigations have focused on exploring alternative battery systems that utilize elements, such as sodium (Na), potassium (K), magnesium (Mg), calcium (Ca), and aluminum (Al). Among these alternatives, aluminum has emerged as an exceptionally promising candidate due to its abundance, low cost, high volumetric and gravimetric capacities, and its inherent safety advantages.^{4–7} The trivalent nature of aluminum enables reversible electro-deposition and electro-stripping at ambient temperature in chloroaluminate-containing ionic liquids or deep eutectic solvents, which makes it a good choice for developing non-flammable and high-energy-density batteries.^{8,9} In the field of Al-battery research, significant attention has been directed toward the investigation of intercalation-type cathodes. These

cathodes include various materials, including graphite,^{4,5,8,10} transition metal sulfides,^{5,10–13} transition metal oxides,^{7,14,15} and conducting polymers¹⁶ – all of which have been subjects of extensive research.^{10,17,18} While the intercalation cathode materials generally exhibit longer life cycles, their moderate energy storage capacities pose a challenge to the creation of aluminum battery systems with high energy densities.³

Rechargeable aluminum–sulfur (Al–S) batteries have garnered significant attention as promising energy storage systems for practical applications owing to their remarkable theoretical capacity (1675 mA h g^{−1}), cost-effectiveness, and environment friendliness. However, the implementation of Al–S batteries in real-world scenarios is hindered by several challenges, despite their numerous advantages. The utilization of immature materials, particularly conversion-type sulfur (S) cathode, in Al–S batteries presents limitations.¹⁹ These limitations encompass low conductivity, sluggish kinetics, significant volume change, the potential occurrence of the “shuttle effect” caused by polysulfide intermediates, and a large voltage hysteresis.^{11,12,20} These drawbacks significantly impede the electrochemical reversibility of the Al–S system and highlight the crucial necessity of fabricating S composite cathodes with precisely engineered designs to enhance cyclic stability. Remarkably, an intriguing observation regarding cathode design is that the same S composite cathode may exhibit considerably inferior electrochemical reversibility in Al–S batteries compared to Li–S batteries.^{21–24}

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In 2016, the first rechargeable aluminum–sulfur (Al–S) battery was introduced, wherein a sulfur cathode and an activated carbon cloth as a conductive matrix were combined with an AlCl_3 electrolyte dissolved in [EMIm]Cl. However, in the past few years, limited research has been conducted on utilizing carbon–sulfur composite cathodes with non-aqueous AlCl_3 : [EMIm]Cl electrolyte. The combination of carbon materials and sulfur enhances the electrochemical performance of the cathode, particularly during the initial cycles.^{16,23–26} Nevertheless, a rapid degradation in capacity has been observed thereafter. The decline in capacity in metal–sulfur batteries can be attributed to various factors, such as the insulating nature of sulfur, its loss into the electrolyte (known as shuttling), and the sluggish solid–solid interconversion between sulfur and its reaction products.^{27,28} Despite the advancement in research, a comprehensive understanding of the operational mechanism and the underlying causes of the rapid capacity decay in Al–S batteries is still lacking, which reflects the state-of-the-art in the current literature.^{8,20} In this research investigation, we aim to elucidate the factors underlying the capacity deterioration observed in rechargeable aluminum–sulfur (Al–S) batteries utilizing graphite nano-powder and sulfur as constituents of the cathode. Through our comprehensive analysis, we have successfully identified and characterized a key phenomenon wherein the interaction between the electrolyte and sulfur species triggers the generation of an obstructive surface passivating layer. This layer comprises insoluble discharge by-products, which engender swift deactivation of the electrodes, thus impairing the overall battery performance during cycling.

2. Experimental

2.1 Chemicals

The chemicals and materials were purchased and used without further purification in this study: Graphite (Sigma-Aldrich, 9.99%), sulfur (Sigma-Aldrich, 99.99%), aluminum chloride (Sigma-Aldrich, 99.99%), 1-ethyl-3-methyl imidazolium chloride (EMImCl; Sigma-Aldrich, 99.99%), *N*-methyl-2-pyrrolidine (NMP; Finar, 99.5%), polyvinylidene fluoride (PVDF, Sigma-Aldrich, 99.9%), acetylene carbon black (Alfa Aesar, 99.9%), orthophosphoric acid (H_3PO_4 , Finar, 99%), Milli-Q water (Sigma, 99%), and isopropanol (Finar, 99.5%). Aluminum foil and stainless steel (SS) mesh were obtained from a local Indian company, RSI Pvt. Ltd. All materials were of analytical grade.

2.2 Synthesis of the S@Gf composite

A non-destructive sublimation-deposition technique was employed for the synthesis of a composite material consisting of sulfur and graphite (referred to as S@Gf). A blend of sulfur and graphite nanopowder (GNP) in a 70:30 weight ratio was homogeneously mixed using an agate mortar and pestle for approximately 3 hours. The resulting mixture exhibited a black color and was transferred to a quartz ampoule under vacuum conditions ($\leq 10^{-2}$ mbar) to prevent exposure to air and contaminants. Subsequently, the sealed mixture was subjected to

heating in a muffle furnace at a heating rate of $5\text{ }^\circ\text{C min}^{-1}$ at a temperature of $150\text{ }^\circ\text{C}$ for approximately 12 hours. After cooling down to room temperature, the resulting composite material that was composed of sulfur and graphite was collected in a powdered form. The formation of the S@Gf composite was confirmed through various analytical techniques, including Fourier-Transform infrared (FT-IR) spectroscopy, X-ray photoelectron spectroscopy (XPS), Raman spectroscopy (RAMAN), X-ray diffraction (XRD), field emission scanning electron microscopy with energy-dispersive X-ray spectroscopy (FESEM-EDAX), and thermal stability was evaluated through thermo-gravimetric analysis (TGA).

2.3 Electrolyte preparation

Anhydrous AlCl_3 powder and EMImCl were utilized as the raw materials for electrolyte preparation. Before use, the materials were purified by preheating them in an evacuated tube furnace at $70\text{ }^\circ\text{C}$ for 8 h to eliminate any residual water. Subsequently, the evacuated materials were transferred to an Argon-filled glove box (with less than 0.5 ppm O_2 and less than 0.5 ppm H_2O) for electrolyte preparation. The electrolyte was prepared by gradually adding AlCl_3 to EMImCl in a molar ratio of 1.3:1 (mol mol^{-1}). The resulting product was a clear, light-yellow liquid formed through an exothermic reaction. The electrolyte was stirred for an hour and stored undisturbed in the glove box.

2.4 Materials characterization

The S@Gf composite material was subjected to various analytical techniques to characterize it structurally and chemically. The X-ray diffraction (XRD) analysis was carried out using a PANalytical Empyrean model equipped with a PIXEL 3D detector (source: Cu $K\alpha$, $\lambda = 1.5404\text{ \AA}$, $2\theta = 5\text{--}80^\circ$) to examine the structural properties. Fourier transform infrared (FT-IR) analysis was performed using a Bruker Alpha-T FT-IR spectrometer with a step resolution of 2 cm^{-1} across the wavenumber range of $400\text{--}4000\text{ cm}^{-1}$ to investigate the effective formation of the S@Gf composite. The chemical environment of the composite cathode material was studied using X-ray photoelectron spectrometry (XPS) on a Kratos Axis Supra instrument. The composite stoichiometry was estimated using TGA on TA Q50 equipment under a nitrogen atmosphere from room temperature to $800\text{ }^\circ\text{C}$ at a scan rate of $10\text{ }^\circ\text{C min}^{-1}$. Ultraviolet-visible spectroscopy (UV-Vis) was performed on a Lambda –950 PerkinElmer for the quantitative determination of analytes (catalyst). The microstructural and elemental distribution of the S@Gf composite was analyzed using FE-SEM on a JEOL7610F FESEM equipped with an OXFORD EDAX detector. Raman spectroscopy (DXR Thermo Fisher Scientific) was used to analyze the vibrational, rotational, and other states in the composite.

2.5 Fabrication of electrodes and cells

The electrochemical performance of the S@Gf active material was evaluated using a standard two-electrode configuration. An Al–S battery was constructed with S@Gf as the cathodic active material and aluminum foil as the anodic active material. The

cathode slurry was prepared by mixing 80 wt% of the active material (graphite/sulfur), 10 wt% of conductive super P, and 10 wt% of PVDF. The mixture was then directly added to an NMP solution with continuous stirring for 10 h to create a viscous slurry. The cathodic slurry was applied onto a pre-cleaned stainless-steel mesh, and the electrode film was dried in a vacuum at 100 °C for 12 hours. A 0.08-mm-thick aluminum foil (0.5 × 0.4 cm) was etched with orthophosphoric acid for 10 minutes and rinsed with pure water and then isopropanol to remove oxide layers. The freshly etched Al-plate was used as the anode in the test cell. The electrode films were cut into rectangular shapes (0.4 cm × 0.5 cm) and assembled into the test cell in an Ar-filled glove box. The electrode film typically weighed 2 mg cm⁻². The electrolyte used was a non-aqueous solution composed of AlCl₃-EMImCl, and a glass fiber was used as the separator.

3. Results and discussion

3.1 Synthesis and structural characterization

3.1.1 Synthesis. A composite active cathode material consisting of sulfur and graphite, referred to as S@Gf, was successfully synthesized using a non-destructive sublimation-deposition technique (as shown in Fig. 1a). Initially, a homogeneous mixture of sulfur and graphite was prepared. The resulting mixture was placed in a sealed tube, where the sulfur sublimed and coordinated with the graphite layer. The weight of the obtained sample was measured to determine the amount of the sulfur incorporated. To confirm the formation and characteristics of the S@Gf material, various analytical techniques were employed. XRD was utilized to examine the crystal structure, while Raman spectroscopy provided information about the molecular composition and bonding. Morphological and elemental analyses were performed with FESEM-EDAX spectroscopy. FT-IR spectroscopy was used to investigate the functional groups present. XPS was employed to study surface chemistry and elemental composition. Furthermore, the thermal stability and sulfur content in the S@Gf material was evaluated using TGA.

3.1.2 X-ray diffraction analysis. Fig. 2a presents the X-ray diffraction (XRD) patterns of pristine sulfur, graphite, and the S@Gf composite. The XRD pattern of pure graphite displays a distinct and intense peak observed at $2\theta = 26.53^\circ$ (JCPDS: 75-1621), which corresponds to the crystallographic plane denoted as (002) in graphite and possesses a d -spacing value of 3.39 Å. Additional broad and low-intensity peaks can be

observed at specific angles: 11.73, 44.25, and 54.4°. Interestingly, the latter two angles closely correspond to diffraction lines at $d = 1.7^\circ$ and 2.04 and 1.7°, respectively, which are indicative of the (101) and (004) planes of graphite. The corresponding d -spacing values are 7.54 Å for the peak at 11.73°. This data suggests that the graphite component of the sample contributes to these peaks. The XRD pattern of pure sulfur displays characteristics of polycrystalline behavior. Multiple peaks with varying intensities can be observed within the 2θ range of 10 to 60°. The most prominent peak occurs at $d = 3.86$, which corresponds to the (222) plane of sulfur, according to the standard data (PDF: 850799) for sulfur with a face-centered orthorhombic structure. Furthermore, a distinct set of sharp peaks with medium-to-low intensities is observed at 2θ angles of 25.83°, 26.71°, 28.71°, 34.18°, 42.81°, 47.80°, and 56.71°. These peaks can be attributed to the (026), (117), (135), (210), (066), and (517) planes of orthorhombic sulfur. Notably, the XRD pattern of the S@Gf composite exhibits nearly identical peaks to those observed in the XRD patterns of pure sulfur and graphite, albeit with slight differences in intensity. These peaks closely resemble the expected Bragg reflections of sulfur with the orthorhombic phase. Moreover, the presence of graphite in the composite is confirmed by the broad peak observed at 26.6° in the XRD pattern of the S@Gf composite. This observation provides evidence for the formation of the S@Gf composite without any accompanying side reactions that could have resulted in the generation of other sulfur-containing compounds. The peaks corresponding to sulfur in the composite match precisely with those of elemental sulfur, indicating the absence of impurities.

3.1.3 FT-IR analysis. The analysis of functional groups within the S@Gf composite was conducted using an FT-IR spectrometer. Fig. 2b presents the FT-IR spectrum of the composite, revealing the presence of various functional groups. The hydroxyl group is discerned by a peak at 3432 cm⁻¹, while the carbonyl group is indicated by a peak at 1734 cm⁻¹. The carbonyl C=O stretching vibration is registered at 1286 cm⁻¹, while the epoxy C-O stretching vibration appears at 1095 cm⁻¹. Stretching vibrations of the -CH₂ group are evident at 2968 cm⁻¹ and 2853 cm⁻¹. Additionally, the C=C stretching vibrations of aromatic rings manifest themselves at 1651 and 1392 cm⁻¹. Sulfur atoms within the composite are present as disulfide bonds, falling within the range of 2855 to 2967 cm⁻¹. Notably, distinct disulfide bond peaks in the FT-IR spectra of the S@Gf composite imply the formation of a hydrogen bond between the carbonyl and sulfonic groups. These functional groups are expected to exert an influence on the electrochemical performance of the composite, and thus, they are important for understanding the material's behavior in battery applications.

3.1.4 Raman analysis. Fig. 2c depicts the Raman spectra of pristine graphite and the S@Gf composite. The pristine graphite spectrum exhibits characteristic peaks known as the D and G bands. Conversely, the S@Gf composite spectrum reveals distinct peaks at various frequencies, indicating the presence of graphite flakes within the composite structure. The D band,

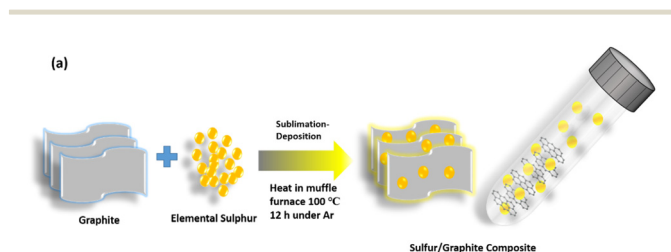


Fig. 1 Synthesis of sulfur/graphite composite by the sublimation-deposition process.

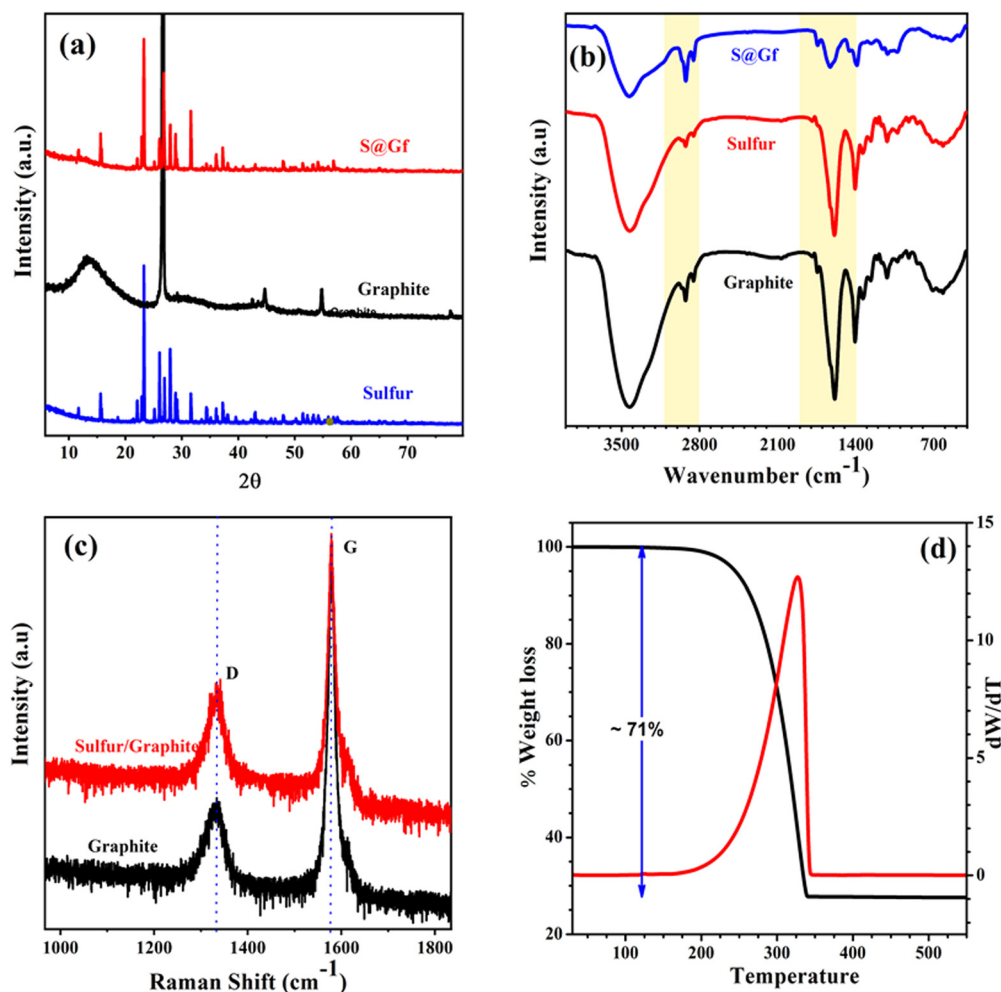


Fig. 2 (a) Stack plots of powder XRD patterns in the scan range of $2\theta = 10$ to 80° , (b) IR spectra, (c) Raman spectra of sulfur, commercially procured graphite and composite S@Gf, and (d) the thermal stability and sulfur composition by TGA in composite S@Gf.

positioned at 1353 cm^{-1} , signifies a prominent peak associated with defect sites and disorders in the carbon framework. In contrast, the G band, situated at 1593 cm^{-1} , corresponds to the graphite-related E_{2g} mode of sp^2 -linked carbon atoms at the zone center. The determination of the I_D/I_G ratio for the S@Gf composite yields a value of 1.51, surpassing that of pristine graphite. This ratio offers valuable insights into the structural modifications that occur during the formation of the S@Gf composite. A comprehensive understanding of the structural characteristics of the composite and the influence of the synthesis process on its properties is attained through Raman spectrum analysis.

3.1.5 Thermo-gravimetric analysis. To quantify sulfur content in the composite material, we employed TGA. The TGA plots presented in Fig. 1d and Fig. S1 (ESI†) depict weight loss from pristine sulfur, graphite, and the S@Gf composite. The measurements were performed under an inert atmosphere (N_2) with a heating rate of $10\text{ }^\circ\text{C min}^{-1}$, spanning a temperature range of $27\text{--}800\text{ }^\circ\text{C}$. Both pristine sulfur and the S@Gf composite demonstrated minimal weight loss up to approximately $165\text{ }^\circ\text{C}$, indicating their stability within this temperature range.

However, beyond this point, both materials exhibited significant weight reduction. Pristine sulfur was entirely consumed at $320\text{ }^\circ\text{C}$ as evident from the absence of any remaining weight (*i.e.*, 0%). Contrarily, pure graphite remained stable up to $600\text{ }^\circ\text{C}$, displaying no weight loss at this temperature. Above $600\text{ }^\circ\text{C}$, graphite underwent oxidation, resulting in a gradual decrease in weight. This oxidation process persisted until approximately $700\text{ }^\circ\text{C}$, signifying nearly complete combustion of the graphite material. The initial phase of rapid weight loss occurred within a broad temperature range of approximately $160\text{--}300\text{ }^\circ\text{C}$, corresponding specifically to the combustion of sulfur within the composite. Subsequently, the weight profile of the composite exhibited a plateau-like dependence on temperature until around $570\text{ }^\circ\text{C}$. Above this temperature, graphite nanoparticles began to oxidize, leading to continuous weight reduction up to $700\text{--}740\text{ }^\circ\text{C}$. At this point, the residual weight of the composite was less than 5%, indicating complete combustion of the constituent materials. Major weight loss from the S@Gf composite occurred between 165 and $325\text{ }^\circ\text{C}$ and amounted to a loss of approximately 71% weight, which serves as an indicator of the sulfur content in the composite. The

remaining 30% was graphite. The high sulfur content in the S@Gf composite can be attributed to the efficient adsorption of a substantial amount of sulfur onto the graphite during the formation of the composite material.

3.1.6 FE-SEM and EDAX analysis. The present study investigates the interaction between sulfur and graphite constituents during the formation of the S@Gf composite using various analytical techniques. Fig. 3(a)–(d) and Fig. S2 (ESI†) present different ways of analysis. The FE-SEM results give valuable insights into the extent of graphite and sulfur mixing and the microscale morphology of the composite material. The absence of aggregates consisting of a single component indicates the successful dispersion of sulfur particles and graphite nanoparticles, resulting in a homogeneous blend of elements interspersed with pores. This unique structure enhances the conductivity of the composite, enabling efficient reaction with aluminum ions during the charge–discharge processes. Furthermore, the EDAX analysis of the S@Gf composite, as illustrated in Fig. 3d, reveals the elemental mapping of carbon and sulfur with weight percentages of 83.4 and 16.3%, respectively. This confirms the successful formation of the S@Gf composite.

3.1.7 XPS analysis. The elemental composition and functional groups present in the S@Gf composite were investigated using XPS. Fig. 4(a)–(c) illustrates the C 1s spectra of the S@Gf composite powder that reveals distinctive peaks at 284.4 and 288.4 eV. The peak at 284.4 eV can be attributed to carbon sp^2 hybridization, indicating the presence of C–C and C=C bonds. Meanwhile, the weaker peak at 288.4 eV signifies the existence of oxygen-containing functional groups, specifically C–O bonds, within the composite. Additionally, the XPS analysis of the S@Gf composite displayed a prominent peak at 285.6 eV in the C 1s spectra, indicating the presence of C–S bonds. This peak confirms the occurrence of chemical bonding between sulfur and graphite in the composite. Notably, the absence of O–C=O bonding suggests that the sulfur nanoparticles fully

functionalize the sites on the edges of the graphite. Moreover, the S 2p XPS spectra of the S@Gf composite exhibited two peaks at 163.7 and 164.6 eV, corresponding to the S 2p_{3/2} and S 2p_{1/2} components, respectively (see Fig. 4(a)–(c)). These peaks can be attributed to either the solid sulfur peak in the composite or the interaction between sulfur and graphite. These XPS findings provide compelling evidence supporting the formation of sulfur-graphite bonding during the synthesis of the S@Gf composite.

3.2 Cyclic voltammetry (CV)

An experimental setup consisting of an aluminum/(S@Gf) test cell was employed to conduct electrochemical investigations. The S@Gf composite served as the cathode, aluminum foil as the anode, and the electrolyte was a solution of AlCl₃ in EMImCl. Cyclic voltammograms were obtained for the Al/S@Gf cell at varying scan rates (1–5 mV s^{−1}) (Fig. 5a and b) within a potential range of 0–2.0 V. The CV curve exhibited distinct anodic and cathodic peaks. The reduction peaks observed at 0.4–0.6 V and 1.2–1.4 V were attributed to different electrochemical reactions between sulfur and aluminum. The oxidation peaks at 1.2 and 1.5 V were associated with the multistep electrochemical reduction of sulfur into higher-order polysulfides (Al₂S_n, where $n = 1–6$). The reduction maxima observed at 0.8–0.3 V may indicate further reduction into Al_xS_y or Al₂S. Additionally, the oxidation signal detected at 1.8 V corresponded to the oxidation process of Al_xS_y. The cyclic voltammogram obtained after 20 cycles showed a significant overlap with the initial cycle, indicating favorable cyclic stability and reversibility. These characteristics can be attributed to the presence of nano-sized charge carriers and the mitigation of electrode volume expansion.

Furthermore, the electrodes underwent CV experiments with various scan rates ranging from 1 to 5 mV s^{−1}. These experiments aimed to ascertain the diffusion coefficient (D). The peak currents corresponding to the primary anodic and cathodic peaks observed for the S@Gf cells with an aluminum reference electrode (RE) and counter electrode (CE) were graphed against the square root of the scan rate. By determining the slopes of these linear regressions and employing the Randles–Sevcik equation [$I_p = (0.4463nFAC(nFD/RT)^{1/2})$, with constants $F = 96485 \text{ C mol}^{-1}$, area (A) = 0.2 cm², temperature (T) = 298 K, and a concentration (C) of 1.3 M (with $n = 1$ for aluminate ions)], we calculated the diffusion coefficient (D) values, which are presented in Table 1.

3.3 Charge–discharge studies

Fig. 6(a) and (b) illustrates the galvanostatic charge–discharge cycles of the aluminum/sulfur-doped graphene foam (Al/S@Gf) cell in the voltage range of 0–2.0 V. When operated at a current density of 300 mA g^{−1} at 50 cycles, the Al/S@Gf cell displayed an initial discharge capacity of 956 mA h g^{−1}. As depicted in Fig. S7a (ESI†), the initial theoretical capacity of the cathode made of S@Gf composite operating at a current density of 300 mA g^{−1} for 3 cycles reached an impressive value of 1281.6 mA h g^{−1}. The discharge profile exhibited a minor

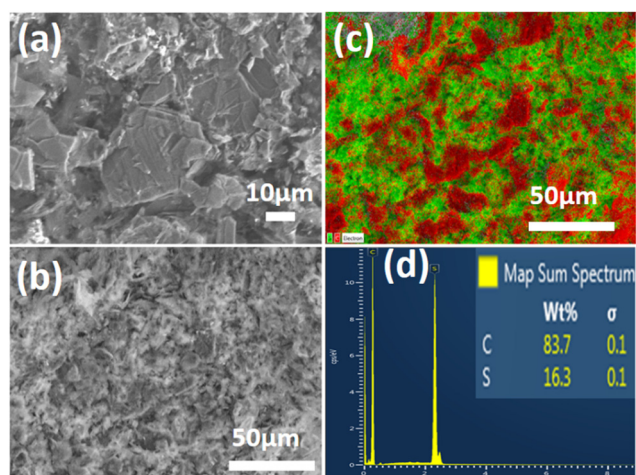


Fig. 3 (a) and (b) FE-SEM images corresponding to the synthesized S@Gf composite, (c) colored layer and elemental mapping, and (d) EDAX spectrum for elemental composition.

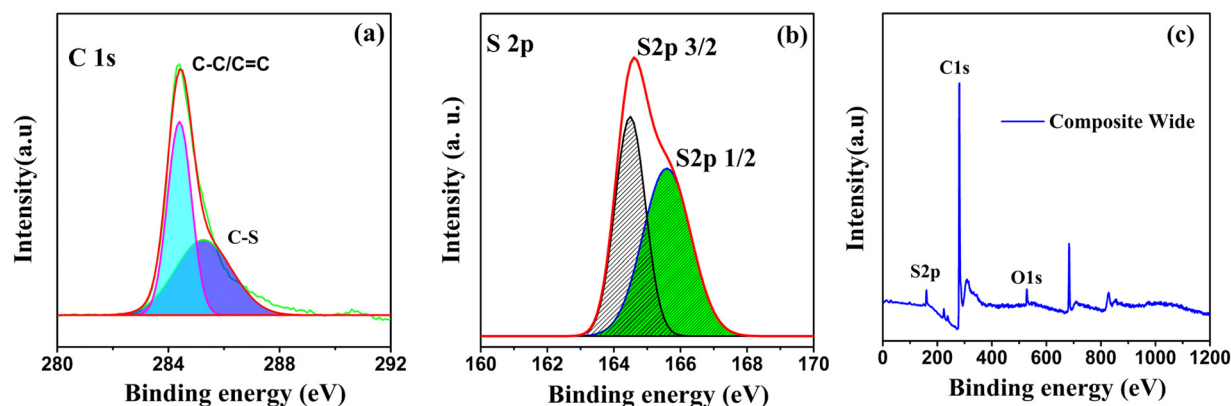


Fig. 4 The XPS spectra of synthesized S@Gf for (a) Carbon C1s, (b) sulfur S2p, and (c) total wide spectra.

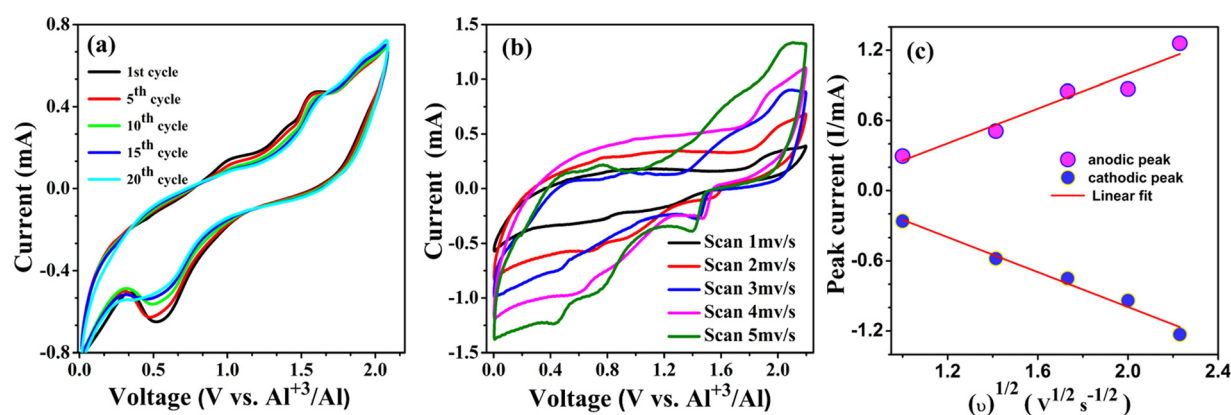


Fig. 5 (a) Cyclic voltammograms were obtained for S@Gf active materials in the eutectic electrolyte against the Al anode at a slew rate of 3 mV s^{-1} in the voltage window of 0.0 to 2.1 V versus Al/Al^{3+} and (b) S@Gf as an active host against Al anodes corresponds to the peak current observed for the respective different slew rates, and (c) the CV results for the cells plotted with respect to the square root of the scan rate.

discharge plateau at 1.5 V, indicating the intercalation of aluminum ions into the S@Gf cathode, which was consistent with the CV behavior. Two distinct plateaus were observed in the potential range, corresponding to the interaction between aluminum and sulfur. The first plateau, occurring at a relatively lower potential (1.8 V), was attributed to the formation of soluble polysulfides in the electrolyte. Conversely, the second plateau observed at a higher potential (1.3 V) indicated slower reaction kinetics compared to the formation of aluminum polysulfides.^{29–31}

The discharge capacity of the S@Gf cathode gradually decreased and stabilized at 800 mA h g^{-1} after from 10 to 30th cycle (Fig. 6b), indicating excellent control over the shuttle effect. Fig. 6b presents a summary of the life cycle properties,

including charge and discharge capacities, as well as the coulombic efficiency of the Al/S@Gf cell. This cell exhibited a coulombic efficiency of 98–100%, corresponding to a capacity retention of 60% indicating considerable control over the shuttle effect. Furthermore, Fig. 6b demonstrates the rate performance of the S@Gf composite electrode, where the average discharge capacities at a current density of 300 mA g^{-1} were 953, 852, 792, 770, 738, 668, and 583 mA h g^{-1} . The cycling up to 100 cycles was continued at the 100th cycle, resulting in drastic fading of the capacity up to 123 mA h g^{-1} . This trend revealed that the S-graphite interaction supports the smooth polysulfide conduction at the 50th cycle. The charge–discharge study at different current densities was analyzed at 200, 300, 400, 400, 600, 700, and 800 mA g^{-1} for the first three cycles. The first cycle was discarded and later two cycles have been reported in Fig. S7 (ESI[†]); each cycle was flat as shown in Fig. S7(a)–(g) (ESI[†]). Cell capacity of less than 221 mA h g^{-1} was not observed. This unique combination of cathode and electrolyte offers a new superior option in Al–S batteries. The present options and the current study are compared in Table S1 (ESI[†]). The unique wrinkled, micro-spherical design of the cathode, along with the presence of internal voids, facilitated

Table 1 The diffusion coefficients for the electrode corresponding to the peak in impedance spectra samples at the indicated significant peaks

Composite	Site	Diffusion c coefficients (D , $\text{cm}^2 \text{ s}^{-1}$)
S@Gf	Anodic	1.19×10^{-10}
S@Gf	Cathodic	1.14×10^{-10}

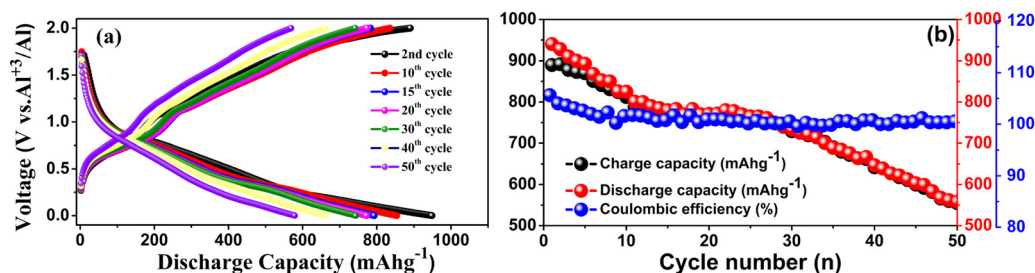


Fig. 6 (a) GCD cycle recorded for S@Gf, respectively, in the potential window 0.0–2.0 V vs. Al/Al³⁺ at current 300 mA g^{−1} and (b) dual y-axis graphs displaying the specific capacities and percent coulombic efficiency as a function of cycle number for the composite electrodes at 300 mA g^{−1} current.

the transport of electrolytes and aluminum ions, resulting in enhanced composite activity, which could account for its exceptional rate performance.

3.4 Post-mortem analysis

Additional validation of the interplay between sulfur and aluminum during galvanostatic charge–discharge (GCD) cycling was attained *via* post-mortem examination by SEM, EDAX spectroscopy, XPS, and UV spectroscopy of the cycled S@Gf cathode. Notably, the analysis revealed the irreversible formation of discharge products. After the completion of 50 GCD cycles, the Al/S@Gf cell was disassembled, and the S@Gf cathode was extracted. The extracted cathode was subjected to sequential rinsing with acetone or ethanol for 3–4 cycles, followed by drying in an oven at 70 °C for 6 hours.

3.5 EDAX and EDAX mapping

The EDAX spectroscopic analysis of the cycled discharge state, *i.e.*, Discharge-S@Gf shown in Fig. 7(a) and (b) and Fig. S2–S6 (ESI[†]), reveals the detection of carbon (C), sulfur (S), and aluminum (Al) elements. The weight percentages of these elements provide compelling evidence for the existence of aluminum in the discharged product. This finding strongly supports the notion of interaction between aluminum and sulfur, leading to the formation of aluminum sulfide compounds during the discharge process.

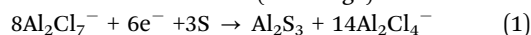
3.6 XPS analysis

To elucidate the reaction mechanism of the Al/S@Gf battery, we analyzed the chemical states of the S@Gf cathode in the discharged state using XPS (Fig. 8(a)–(d)). The C 1s appeared at 281.5 and 283.5 eV revealing the interaction with S 2p (Fig. 4a). The S 2p spectrum of the pristine S@Gf cathode from XPS analysis, as shown in Fig. 4b, exhibited characteristic peaks corresponding to the S8 structure. The S 2p_{1/2} and S 2p_{3/2} peaks were observed at binding energies of 165.3 and 164.6 eV, respectively.

Upon discharge, a slight shift of 0.05 eV toward lower binding energies was observed in the S@Gf cathode's XPS spectrum. This shift indicated the presence of larger sulfur-containing species, predominantly Al₂S₃, resulting from the predominant transformation of sulfur during the discharge process. The coexistence of elemental sulfur and partially

reduced sulfur further confirmed an incomplete reduction reaction of the sulfur cathode. Deconvolution analysis of the S 2p²⁷ peak during discharge revealed distinct peaks at binding energies of 164.2 eV (Fig. 4b), 163.3 eV (S₂^{4−}), 162.5 eV (partially reduced S[−]), and 161.2 eV (fully reduced S^{2−}), respectively.^{27,28} These findings indicate that the sulfur-positive electrode undergoes reduction to form aluminum-sulfide (Al₂S₃) during the discharge process, which is then oxidized back to elemental sulfur during the charging process. The electrochemical reaction mechanism of the Al/S@Gf battery can be described by the following equation:

Positive electrode (discharge):



Negative electrode (charge):

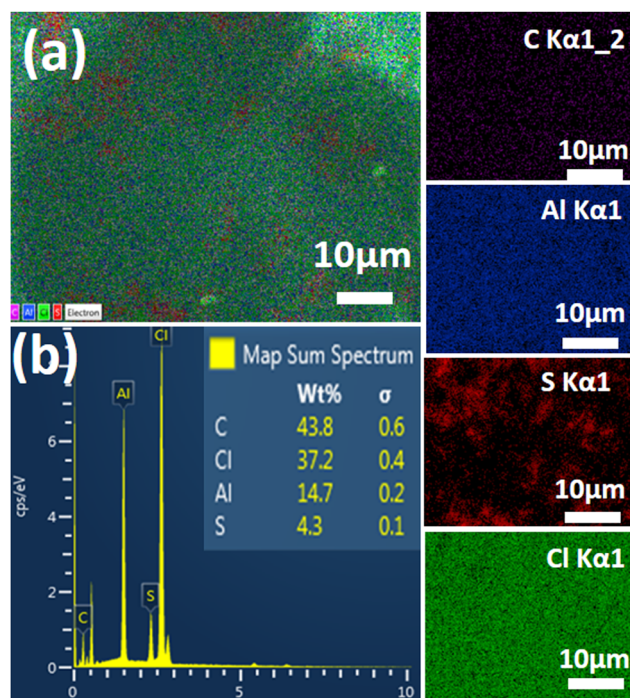
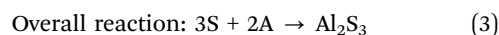
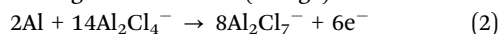


Fig. 7 (a) Elemental mapping shown in color, and (b) EDAX images of S@Gf the cycled electrodes over 50 GCD cycles.

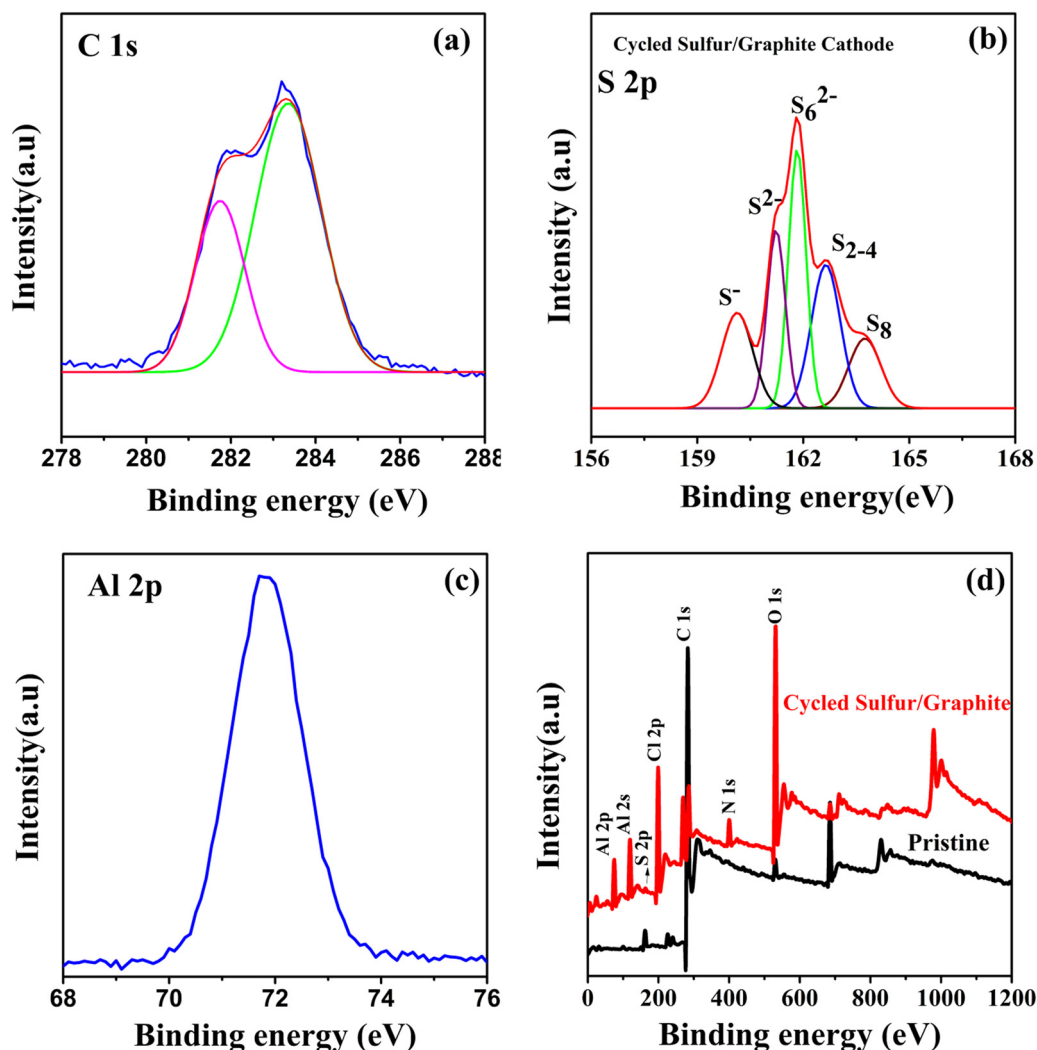


Fig. 8 The XPS spectra of the cycled S@Gf composite electrode from 50 GCD cycles (a) carbon, C 1s, (b) sulfur, S 2p, (c) aluminum, Al 2p, and (d) wide XPS spectra of the cycled S@Gf composite electrode.

3.7 UV-Vis analysis

Unlike Li-S or Mg-S batteries,^{27,32} to examine the patterns of charge and discharge and identify intermediate products in the sulfur cathode, we conducted studies utilizing UV-Vis and X-ray spectroscopies. The XPS spectra for discharged or charged cathodes obtained from Al-S cells, along with the findings from the UV-Vis spectroscopic analysis, are presented in Fig. 9. To obtain liquid samples, the discharged or charged cathodes, as well as the separators, were submerged in dichloromethane (DCM) as the solvent for 48 hours. According to previously published literature,^{27,33} the absorbance peak observed around 280 nm in Fig. 5a is attributable to dissolved elemental sulfur. Except for the fresh sulfur cathode, all discharged or charged cathodes exhibit a characteristic absorbance peak at 320–360 nm, corresponding to S_n^{2-} ($4 \leq n \leq 6$) species.^{21,23} However, the intensity of these S_n^{2-} ($4 \leq n \leq 6$) peaks vary significantly depending on the state of charge or discharge. The S_n^{2-} ($4 \leq n \leq 6$) peaks are much stronger in the discharged (0.1 V) and charged (to 2.1 V) cathodes.

Interestingly, these S_n^{2-} ($4 \leq n \leq 6$) peaks remain visible even in the fully discharged (to 0.1 V) and fully charged (to 2.1 V) cathodes, indicating the presence of a small quantity of Al_2S_3 ($4 \leq n \leq 6$) species within the cathode, even at high magnitudes of discharge or charge.

3.8 Electrochemical impedance studies

An Al cell with a sulfur@Gfaphite (S@Gf) cathode was subjected to impedance spectroscopy (Fig. 10) before and during cycling. The initial discharge capacity of the cathode was 953 mA h g^{-1} , and it remained above 580 mA h g^{-1} consistently throughout 50 cycles. Graphite ribbon-like structures create a stable interface between the liquid electrolyte and the aluminum polysulfides within the cathode, allowing the propagation of mostly aluminum ions while limiting capacity loss.

In the high-frequency region, the Nyquist plots adopted the shape of a semicircle, while then in the low-frequency region, they adopted an s-shaped line. The semicircle represents a parallel connection between the capacitor and the resistor,

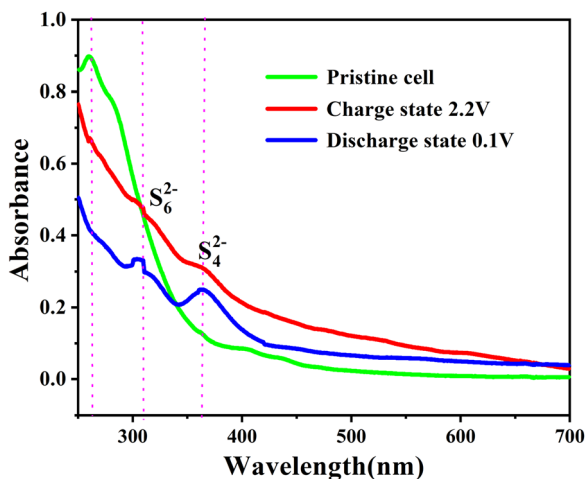


Fig. 9 Spectra from UV analysis of the pristine sample, as well as the cell electrolyte (DCM solvent used for UV experiment) in the charge and discharge states.

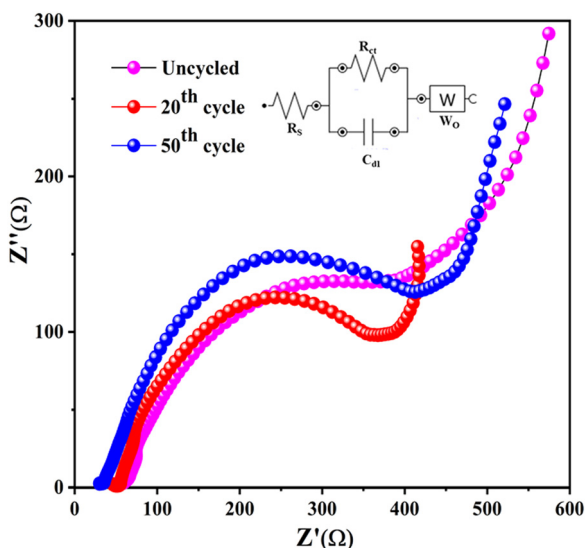


Fig. 10 The Nyquist plots; the circuit diagram for the test cell is shown in the inset: plots are for synthesized S@Gf and cycled S@Gf cathodes in the frequency range of 20 kHz to 100 mHz with an amplitude of ± 10 mV.

with the former denoting double-layer charging and the latter indicating charge transfer resistance. The low-frequency straight line is consistent with a diffusion-controlled mechanism, especially the diffusion of aluminum ions throughout the bulk of the electrode. Fig. 10 shows that after 50 cycles, the charge transfer resistance (R_{ct}) of the S@Gf cathode was significantly reduced from its initial value of about 43 to 33 Ω . The capacity of the S@Gf cell dropped from 953 mA h g^{-1} after the first cycle to 580 mA h g^{-1} after 50 cycles; this decline in R_{ct} may be attributed to the loss of active material. R_{ct} went up to 80 after the first cycle, most likely because a solid electrolyte interphase (SEI) had formed. By the 20th cycle, the capacity had risen to 767 mA h g^{-1} , but R_{ct} remained the same throughout. When polysulfides dissolve in the electrolyte and migrate to the

anode, the active material is lost, which accounts for the small drop in capacity during cycling.

The thickness of the SEI layer increased with cycling, resulting in increased resistance to the flow of aluminum ions towards the cathode material. By the end of the 50th cycle, the interfacial resistance reached 33 Ω , coinciding with a decrease in discharge capacity to 580 mA h g^{-1} . Subsequently, R_{ct} experienced a significant drop, indicating the collapse of the SEI layer and reduced loading of the active material content on the current collector, leading to gradual capacity degradation. The barrier to the movement of aluminum ions toward the cathode increased as the thickness of the SEI layer grew with repeated cycling. After 50 cycles, the interfacial resistance increased to 33 Ω , and the discharge capacity dropped to 580 mA h g^{-1} . As a result, R_{ct} dropped dramatically, revealing the termination of the SEI layer and diminished active material content on the current collector; both of these cell components contributed to capacity deterioration over time. From cycle 11 through 50, the cell maintained a coulombic efficiency between 98 and 100%. Aluminum ions combine with sulfur (S_8) to generate aluminum sulfide (Al_2S_3) in a reversible discharge process, and the quantity of aluminum ions released during charging is unaffected by capacity loss and the development of the SEI. This indicates that structural and interfacial stabilization was generated in the sulfur electrode by the incorporation of graphite, and it survived the removal of the active material from the current collector.

4. Conclusion

In conclusion, a new approach was developed for the prohibition of the dissolution of polysulfides because of the effective pairing of the sublimated sulfur within graphite and a suitable electrolyte, *i.e.*, (EMImCl): AlCl_3 . The Al/EMImCl: AlCl_3 /S@Gf battery exhibited remarkable performance in terms of rate capability and cycle life, delivering a high capacity of 1281 mA h g^{-1} at a current density of 200 mA g^{-1} with 100% coulombic efficiency (CE); it was stable at 800 mA g^{-1} withdrawing current. The electrochemical properties exhibited by the composite of sulfur within a graphite framework (S@Gf) were attributed to its distinctive structure characterized by wrinkled micro-spheres. Graphite serves as a host matrix that confines sulfur, affording adequate space for accommodating both Al-S ions and polysulfides. The properties of the eutectic electrolyte mixture minimize the dissolution of active materials and suppress the dissolution of polysulfide ions. Furthermore, this investigation demonstrates the uniform deposition of sulfur particles onto graphite *via* sublimation-deposition, which enhances electrical conductivity. By pairing of cost-effective and scalable electrolyte electrode pairs, aluminum-sulfur batteries could achieve exceptional performance levels.

Author contributions

Mr Mohan Gorle: conceptualization, methodology, investigation, formal analysis, visualization, writing – original draft and

review. Dr Vatsala Rani Jeti: supervision, conceptualization, funding acquisition, resources, writing – review and editing. All authors have approved the final version of the manuscript.

Conflicts of interest

There are no conflicts to declare.

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